

Discrete Mathematics

CSE 4203

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4.3 Primes and Greatest Common Divisors

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- 4.1 Divisibility and Modular Arithmetic
- 4.2 Integer Representations and Algorithms
- 4.3 Primes and Greatest Common Divisors
- 4.4 Solving Congruences
- 4.5 Applications of Congruences
- 4.6 Cryptography

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Applications:

- Number theory, cryptography, and computational mathematics.

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Other Key Topics:

- **Fermat's Little Theorem:** If p prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

- **Pseudoprimes and Carmichael Numbers:** Composite numbers behaving like primes.
- **Discrete Logarithms and Primitive Roots:** Basis of cryptographic security.

Linear Congruences and Inverses

Definition: A congruence of the form:

$$ax \equiv b \pmod{m},$$

where m is a positive integer, and a, b are integers, is called a **linear congruence**.

Goal: Find all integers x satisfying the congruence.

Key Idea:

- If there exists a^{-1} such that:

$$aa^{-1} \equiv 1 \pmod{m},$$

then the solution is:

$$x \equiv a^{-1}b \pmod{m}.$$

Theorem 1: If $\gcd(a, m) = 1$ and $m > 1$, then:

- An inverse of a modulo m exists.
- This inverse is **unique modulo m** .

Theorem: Existence and Uniqueness of Modular Inverse

Theorem 1

If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists. Moreover, this inverse is unique modulo m .

Proof: Since $\gcd(a, m) = 1$, by Bézout's Theorem there exist integers s and t such that

$$sa + tm = 1.$$

Taking both sides modulo m :

$$sa \equiv 1 \pmod{m}.$$

Thus, s is an inverse of a modulo m .

Uniqueness: Suppose x and y are both inverses of a modulo m . Then

$$ax \equiv 1 \pmod{m} \quad \text{and} \quad ay \equiv 1 \pmod{m}.$$

Subtracting gives

$$a(x - y) \equiv 0 \pmod{m}.$$

Since $\gcd(a, m) = 1$, we have $x \equiv y \pmod{m}$.

Modular Inverse: Theorem and Example

Example: Find inverse of 7 mod 26

Step 1: Apply Extended Euclidean Algorithm

$$26 = 3 \cdot 7 + 5, \quad 7 = 1 \cdot 5 + 2, \quad 5 = 2 \cdot 2 + 1$$

Step 2: Work Backwards

$$1 = 5 - 2 \cdot 2$$

$$1 = 5 - 2(7 - 1 \cdot 5) = 3 \cdot 5 - 2 \cdot 7$$

$$1 = 3(26 - 3 \cdot 7) - 2 \cdot 7 = 3 \cdot 26 - 11 \cdot 7$$

Result:

Inverse of 7 mod 26 is $-11 \equiv 15 \pmod{26}$.

Example: Solving a Linear Congruence

Problem: Find all solutions of

$$3x \equiv 4 \pmod{7}.$$

Solution:

- Inverse of 3 mod 7 is -2 (since $3 \cdot (-2) \equiv 1 \pmod{7}$).
- Multiply both sides by -2 :

$$-2 \cdot 3x \equiv -2 \cdot 4 \pmod{7}.$$

- Compute:

$$(-6)x \equiv -8 \pmod{7} \implies x \equiv 6 \pmod{7}.$$

Check:

$$3 \cdot 6 = 18 \equiv 4 \pmod{7}.$$

Solution Set:

$$x \equiv 6 \pmod{7} \implies \{\dots, -8, -1, 6, 13, 20, \dots\}.$$

Theorem 2

Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers greater than 1 and $a_1, a_2, \dots, a_n$ be integers. The system

$$x \equiv a_1 \pmod{m_1},$$

$$x \equiv a_2 \pmod{m_2},$$

$$\vdots$$

$$x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m = m_1 m_2 \dots m_n$.

Constructing a Solution:

- 1 Let $m = m_1 m_2 \dots m_n$.
- 2 For each $k = 1, \dots, n$, define:

$$M_k = \frac{m}{m_k}.$$

- 3 Since $\gcd(m_k, M_k) = 1$, find y_k such that:

$$M_k y_k \equiv 1 \pmod{m_k}.$$

(This is the modular inverse of $M_k \bmod m_k$.)

- 4 Compute:

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + \dots + a_n M_n y_n.$$

- 5 Take $x \bmod m$ for the unique solution.

Why This Construction Works

Check congruences:

- For k -th congruence:

$$x = \sum_{j=1}^n a_j M_j y_j.$$

- If $j \neq k$, then $M_j \equiv 0 \pmod{m_k}$.
- So:

$$x \equiv a_k M_k y_k \pmod{m_k}.$$

- Since $M_k y_k \equiv 1 \pmod{m_k}$:

$$x \equiv a_k \pmod{m_k}.$$

Uniqueness: If x_1, x_2 satisfy all congruences, then $x_1 \equiv x_2 \pmod{m}$ because $m_1, m_2, \dots, m_n$ are pairwise relatively prime.

Chinese Remainder Theorem: Example

Problem: Find x such that:

$$\begin{cases} x \equiv 2 \pmod{3}, \\ x \equiv 3 \pmod{5}, \\ x \equiv 2 \pmod{7}. \end{cases}$$

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Step 1: Compute M :

$$m_1 = 3, m_2 = 5, m_3 = 7, M = m_1 \cdot m_2 \cdot m_3 = 105.$$

Step 2: Compute partial moduli:

$$M_1 = \frac{M}{m_1} = 35, M_2 = 21, M_3 = 15.$$

Chinese Remainder Theorem: Solution

Step 3: Compute inverses:

$$y_1 = 35^{-1} \pmod{3} = 2, \quad y_2 = 21^{-1} \pmod{5} = 1, \quad y_3 = 15^{-1} \pmod{7} = 1.$$

Step 4: Compute x :

$$x = (2 \cdot 35 \cdot 2) + (3 \cdot 21 \cdot 1) + (2 \cdot 15 \cdot 1) = 140 + 63 + 30 = 233.$$

Final Answer:

$$x \equiv 233 \pmod{105} \implies x = 23.$$

THEOREM 3 (Fermat's Little Theorem):

If p is prime and a is an integer not divisible by p , then:

$$a^{p-1} \equiv 1 \pmod{p}.$$

Furthermore:

$$a^p \equiv a \pmod{p}.$$

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Write $222 = 22 \cdot 10 + 2$:

$$7^{222} = 7^{22 \cdot 10 + 2} = (7^{10})^{22} \cdot 7^2 \equiv (1)^{22} \cdot 49 \equiv 49 \pmod{11}.$$

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Finally:

$$49 \bmod 11 = 5.$$

Answer: $7^{222} \bmod 11 = \boxed{5}$.

Pseudoprime Definition and Example

Definition 1: Let b be a positive integer. If n is a composite positive integer, and

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then n is called a **pseudoprime to the base b** .

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Example: Consider $b = 2$ and $n = 341$.

$$341 = 11 \cdot 31 \text{ (composite)}$$

Compute:

$$2^{340} \bmod 341 = 1.$$

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Example: Consider $b = 2$ and $n = 341$.

$$341 = 11 \cdot 31 \text{ (composite)}$$

Compute:

$$2^{340} \bmod 341 = 1.$$

Therefore, 341 is a **pseudoprime to base 2**.

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Example:

$$n = 561 = 3 \times 11 \times 17.$$

561 is composite, but

$$b^{560} \equiv 1 \pmod{561}$$

for all b such that $\gcd(b, 561) = 1$.

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561 is composite, but

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for all b such that $\gcd(b, 561) = 1$. Therefore, 561 is a **Carmichael number**.

Definition 3: A **primitive root modulo a prime** p is an integer $r \in \mathbb{Z}_p$ such that every nonzero element of $\mathbb{Z}_p$ is a power of r .

Primitive Roots Modulo a Prime

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Example 12: Determine whether 2 and 3 are primitive roots modulo 11.

For $r = 2$:

$$2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 5, 2^5 = 10,$$

$$2^6 = 9, 2^7 = 7, 2^8 = 3, 2^9 = 6, 2^{10} = 1$$

All nonzero elements appear $\rightarrow$ 2 is a primitive root.

For $r = 3$:

$$3^1 = 3, 3^2 = 9, 3^3 = 5, 3^4 = 4, 3^5 = 1$$

Not all elements appear $\rightarrow$ 3 is **not** a primitive root.

Definition: Discrete Logarithm

Definition 4: Suppose p is a prime, r is a **primitive root** modulo p , and a is an integer with

$$1 \leq a \leq p - 1.$$

If

$$r^e \equiv a \pmod{p} \quad \text{and} \quad 0 \leq e \leq p - 1,$$

then e is the **discrete logarithm** of a modulo p to the base r .

We write: $\log_r a = e$ (modulus p is understood).

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Key Idea: Discrete logarithm is the inverse of modular exponentiation.

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Solution: From earlier computations:

$$2^8 \equiv 3 \pmod{11}, \quad 2^4 \equiv 5 \pmod{11}.$$

Hence:

$$\log_2 3 = 8, \quad \log_2 5 = 4 \pmod{11}.$$

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Hence:

$$\log_2 3 = 8, \quad \log_2 5 = 4 \pmod{11}.$$

Interpretation:

- $2^8 \bmod 11 = 3$
- $2^4 \bmod 11 = 5$